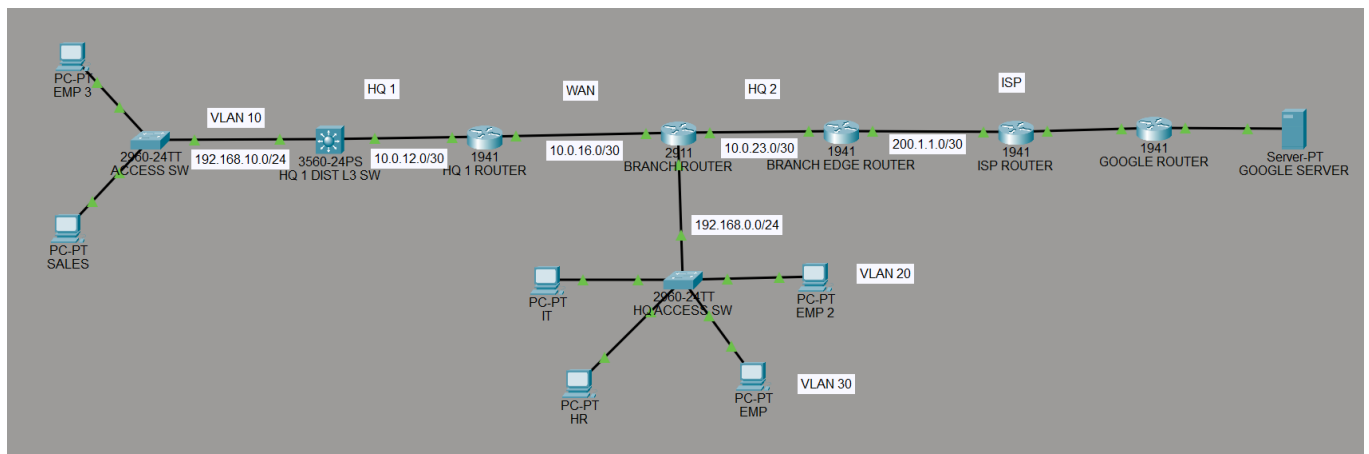


ENTERPRISE PROJECT 2

HQ–Branch WAN Network with OSPF + Static Default + NAT OVERLOAD

DIAGRAM:



DESCRIPTION:

This project simulates an enterprise network with multiple VLANs, inter-VLAN routing, and WAN connectivity using OSPF between a headquarters and a branch office. An edge router performs NAT to translate private IP addresses into public IPs so internal users can access external networks. A public web server is included to represent an internet service like Google. In the lab, static routing is used between the ISP router and the Google router to keep the design simple.

In real-world networks, traffic to services like Google would pass through many ISP networks using BGP (Border Gateway Protocol), not static routes. Large providers use dynamic routing for scalability and redundancy, and often use large public IP pools with advanced NAT and load-balancing at their edge routers. The lab uses a single ISP device to represent the internet, but it still demonstrates real concepts like OSPF routing, NAT and end-to-end connectivity.

STEP 1: Create VLANs, Access Ports & Trunk:

Create Vlan 10, Vlan 20 and Vlan 30 on all switches.

Enable access port for each PC port based on VLAN for relevant Switches.

allow trunks on required paths.

STEP 2: Configure HQ1 access switch (2960):

```
vlan 10
name HQ_USERS

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 10

interface fa0/24
switchport mode trunk
switchport trunk allowed vlan 10
```

STEP 3 : Configure HQ1 L3 switch (3560):

```
ip routing

vlan 10
name HQ_USERS

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown

interface fa0/24
switchport mode trunk
switchport trunk allowed vlan 10

interface g0/1
no switchport
ip address 10.0.12.1 255.255.255.252
no shutdown

router ospf 1
network 192.168.10.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
```

STEP 4: Configure HQ1 router:

```
interface g0/0  
ip address 10.0.12.2 255.255.255.252  
no shutdown
```

```
int g0/1  
ip address 10.0.16.1 255.255.255.252  
no shutdown
```

```
router ospf 1  
network 10.0.12.0 0.0.0.3 area 0  
network 10.0.16.0 0.0.0.3 area 0
```

STEP 5: Configure Branch access switch (2960):

```
vlan 20  
name BRANCH_EMP
```

```
vlan 30  
name BRANCH_HR
```

```
interface fa0/1  
switchport mode access  
switchport access vlan 20
```

```
interface fa0/2  
switchport mode access  
switchport access vlan 30
```

```
interface g0/2  
switchport mode trunk  
switchport trunk allowed vlan 20,30
```

STEP 6 : Configure branch router:

```
interface g0/0  
ip address 10.0.16.2 255.255.255.252  
no shutdown
```

```
interface g0/1  
ip address 10.0.23.1 255.255.255.252  
no shutdown
```

```
interface g0/2
no shutdown

interface g0/2.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

interface g0/2.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0

ip route 0.0.0.0 0.0.0.0 10.0.23.2 (static default route)
```

```
router ospf 1

network 10.0.16.0 0.0.0.3 area 0

network 192.168.20.0 0.0.0.255 area 0

network 192.168.30.0 0.0.0.255 area 0

default-information originate
```

STEP 7 : Configure branch edge router (NAT + ISP Link):

```
interface g0/0
ip address 10.0.23.2 255.255.255.252
ip nat inside
no shutdown

interface g0/1
ip address 200.1.1.2 255.255.255.252
ip nat outside
no shutdown

ip route 0.0.0.0 0.0.0.0 200.1.1.1 (default route)

access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface g0/1 overload
ip route 192.168.0.0 255.255.0.0 10.0.23.1 (static route back)
```

STEP 8: Configure ISP router:

```
interface g0/1
ip address 200.1.1.1 255.255.255.252
no shutdown

interface g0/2
ip address 8.8.8.1 255.255.255.252

no shutdown

ip route 200.1.1.0 255.255.255.252 200.1.1.2      (back to enterprise)

ip route 8.8.4.0 255.255.255.0 8.8.8.2          (route to google static)
```

STEP 9: Configure Google router:

```
interface g0/0
ip address 8.8.8.2 255.255.255.252
no shutdown

interface g0/1
ip address 8.8.4.1 255.255.255.0
no shutdown

ip route 0.0.0.0 0.0.0.0 8.8.8.1 (default route)
```

STEP 10: PCs and server configurations:

PC	VLAN	IP Address	Gateway/CIDR
SALES, EMP3	10	192.168.10.10, 192.168.10.11	192.168.10.1/24
IT, EMP2	20	192.168.20.10, 192.168.20.11	192.168.20.1/24
HR, EMP	30	192.168.30.10, 192.168.30.11	192.168.30.1/24
SERVER	-	8.8.4.10	8.8.4.1/24

STEP 11: Verification and Ping test:

Verification:

```
show ip route
show ip route ospf
show ip route 0.0.0.0
show ip ospf neighbor
show ip ospf interface brief
show ip ospf database
show ip interface brief
show vlan brief
show interfaces trunk
show ip nat translations
show ip nat statistics
```

Ping test:

Ping result from VLAN 10(EMP PC) in HQ 1 pc to VLAN 20 in Branch and Google Server:

```
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time=1ms TTL=125
Reply from 192.168.20.11: bytes=32 time<1ms TTL=125
Reply from 192.168.20.11: bytes=32 time<1ms TTL=125
Reply from 192.168.20.11: bytes=32 time=11ms TTL=125

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 8.8.4.10

Pinging 8.8.4.10 with 32 bytes of data:

Reply from 8.8.4.10: bytes=32 time=10ms TTL=122
Reply from 8.8.4.10: bytes=32 time=19ms TTL=122
Reply from 8.8.4.10: bytes=32 time=11ms TTL=122
Reply from 8.8.4.10: bytes=32 time=11ms TTL=122

Ping statistics for 8.8.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 19ms, Average = 12ms
```

Ping test from HQ 1 to branch router:

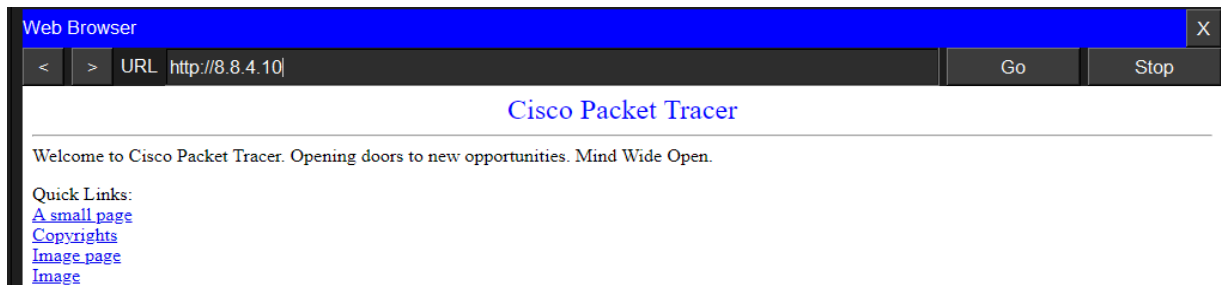
```
Router>ping 10.0.16.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.16.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

Nat translation in branch Edge router:

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  200.1.1.2:1025      192.168.10.10:1025 8.8.4.10:80        8.8.4.10:80
Router#
```

Web browsing Test from EMP 3 PC(VLAN 10):



NOTE: IRL, HQ 1 & BRANCH is connected between internet. In our topology we focus mainly on internal routing concepts, so its directly connected for simplicity.